

ARK-445 — RESULTS

Tri-State Authorization Discrimination (ALLOW / HOLD / DENY)

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Backend: `ibm_marrakesh` (156-qubit Heron r2) · Instance: `open-instance`

Verdict: FAIL (preregistered decision rule, Field 7)

The experiment executed cleanly on hardware with an in-situ SPAM baseline far within ceiling, so the result is a **valid FAIL**, not INDETERMINATE. It fails on **exactly one** of the five preregistered PASS criteria: the reset-based **confusion/replay DENY arm (arm9)** leaked `L_conf_corrected = 0.0289`, above the strict `≤ 0.02` DENY ceiling.

The **core tri-state discrimination is strong and clearly demonstrated**: ALLOW, HOLD, and DENY map to well-separated payload-execution probabilities with a tri-state margin `Δ_H = 0.463`, comfortably above the required `0.30`. The failure is confined to one adversarial DENY arm whose mechanism (mid-circuit `reset` then re-prepare) is sensitive to hardware reset infidelity. **No post-data rescue is applied** (Field 19): the preregistered rule requires ALL four DENY arms to clear the ceiling, so the overall verdict is FAIL.

Provenance (strict ordering — Field 27)

Step	Artifact	Job ID	Commit-before
LOCK	MANIFEST.txt (SHA-256), prereg + 6 code files	—	b0b56b0
Qubit selection	selected_qubits.json	—	893b147
SPAM kill-gate	spam_results.json	d9coda9htsac739c1d- jg	cc745e5 (before principal)
Principal submit	principal_job_id.txt , principal_job_meta.json	d9codk9htsac739c1- dug	9b79d0a (before re- trieval)
Raw results	raw_results.json	—	57561f1
Analysis	proofrecord.json , plots	—	588d61a

- **Qubits (frozen before any job):** Q_A = 2 (RE = 0.0032), Q_P = 1 (RE = 0.0022), connected, sum_RE = 0.0054 , initial_layout = [Q_A, Q_P] = [2, 1] .
- **Shots:** 8,192 per arm. **Principal job:** arms 1–9. **SPAM gate:** arm10 (idle).
- **Transpile:** optimization_level=3 , seed_transpiler=445 , no dynamical decoupling.
- **Primary endpoint:** RAW counts, no readout mitigation (DENY arms SPAM-corrected only).

SPAM kill-gate (arm10, committed before principal — PASSED)

Metric	Value	Ceiling	Pass
SPAM_A (Q_A idle)	0.0007	0.02	✓
SPAM_P (Q_P idle)	0.0011	0.02	✓
SPAM_drift A–P	0.0004	0.005	✓

Primary metrics (Field 6)

ALLOW arms — payload should execute (floor 0.90)

Arm	Metric	P(Q_P=1) raw	≥ 0.90
arm1 allow_standard	S_A	0.9805	✓
arm5 allow_alt	S_A_alt	0.9771	✓
arm7 allow_reverified (1μs delay)	S_A_rev	0.9626	✓
S_A_min		0.9626	✓

HOLD arms — should be intermediate [0.40, 0.60]

Arm	Metric	P(Q_P=1)	in band
arm3 hold_plus (+))	H_plus	0.4946	✓
arm4 hold_minus (-))	H_minus	0.4918	✓
H_min / H_max		0.4918 / 0.4946	✓
HOLD symmetry H_plus-H_minus	I_H	0.0028	(< 0.05)

DENY arms — payload must be withheld (ceiling 0.02, SPAM-corrected)

Arm	Metric	raw	corrected	≤ 0.02
arm2 deny_standard	L_D	0.0040	0.0029	✓
arm6 deny_alt	L_D_alt	0.0029	0.0018	✓
arm8 deny_expired (1μs delay)	L_D_exp	0.0022	0.0011	✓
arm9 confu- sion_replay	L_conf	0.0300	0.0289	✗ breach
L_D_max (cor- rected)			0.0289	✗

Tri-state discrimination margin

- $\Delta_H = \min(S_{A_min} - H_{max}, H_{min} - L_{D_max}) = \min(0.9626 - 0.4946, 0.4918 - 0.0289)$
- $\Delta_H = \min(0.4680, 0.4629) = **0.4629** \geq 0.30$ ✓

PASS/FAIL scorecard (Field 7)

#	Criterion	Threshold	Observed	Result
1	ALLOW floor	$S_{A_min} \geq 0.90$	0.9626	✓
2	DENY ceiling (all 4)	each ≤ 0.02	arm9 = 0.0289	✗
3	HOLD range (both)	[0.40, 0.60]	[0.4918, 0.4946]	✓
4	Tri-state margin	$\Delta_H \geq 0.30$	0.4629	✓
5	SPAM kill-gate	≤ 0.02 / qubit	0.0007, 0.0011	✓

Four of five criteria pass. Criterion 2 fails on arm9 alone → overall **FAIL**.

Secondary hypotheses (Field 3)

- **H2a (HOLD basis symmetry):** SUPPORTED. $H_{plus} = 0.4946$, $H_{minus} = 0.4918$, $I_H = 0.0028$ (< 0.05); 95% CIs overlap. $|+\rangle$ and $|-\rangle$ encode indistinguishable ambiguity.
- **H2b (Reverification escape):** SUPPORTED. $S_{A_rev} = 0.9626 \geq 0.90$ — the reverified ALLOW arm (1 μ s delay) stays high, so HOLD ≈ 0.5 is a genuine encoding, **not** a decoherence artifact.
- **H2c (Confusion falls to DENY):** PARTIALLY SUPPORTED (directional, not to threshold). $L_{conf_corrected} = 0.0289$ sits far closer to the DENY floor (≈ 0.001 – 0.003) than to the HOLD band (≈ 0.49) — the tamper/replay clearly collapses **toward** DENY, not HOLD — but it does not meet the strict ≤ 0.02 ceiling, which is the single cause of the FAIL verdict.
- **H2d (SPAM-drift bound):** SUPPORTED. $SPAM_A = 0.0007$, $SPAM_P = 0.0011$, drift = $0.0004 \leq 0.005$.

Interpretation

Three authorization states are **metrologically separable** on this 2-qubit setup: ALLOW (≈ 0.97 – 0.98), HOLD (≈ 0.49), and DENY (≈ 0.001 – 0.003) occupy cleanly distinct payload-execution bands, with $\Delta_H = 0.46$ well above the preregistered 0.30 . The HOLD state — arising from textbook measurement-induced collapse of a superposed authorizer ($|+\rangle/|-\rangle$) — is stable, basis-independent, and survives a 1 μ s re-verification delay.

The experiment nonetheless **FAILS its preregistered composite criterion** because the adversarial confusion/replay arm (arm9), which uses a mid-circuit `reset` followed by re-preparation, leaks `0.0289` after SPAM correction — above the strict `0.02` DENY ceiling. The most likely mechanism is **reset infidelity** on this Heron qubit: an imperfect reset leaves residual $|1\rangle$ population that occasionally records an approval ($ca = 1$) and fires the payload. This is a real hardware-level leak on the tamper channel, correctly flagged by the preregistered rule. Per Field 19 (no post-data rescue), the ceiling is not relaxed and the arm is not redefined; the verdict stands as FAIL.

What this does and does not mean

- The tri-state **discrimination** claim (ALLOW vs HOLD vs DENY, H1's separation) is strongly and cleanly demonstrated ($\Delta_H = 0.46$).
- The **fail-closed robustness** of the tamper/replay path to the strict 0.02 boundary is **not** met on this backend/calibration; the confusion arm leaks $\sim 2.9\%$.
- This is a **metrological characterization**, not new physics (superposition $\rightarrow$ probabilistic measurement is textbook QM) and **not** a cryptographic guarantee. Findings apply only to qubits $\{2, 1\}$ on `ibm_marrakesh` at this calibration.

Honest scope / provenance notes

- Commits are ordered but **not** GPG-signed; ordering is evidenced by commit timestamps and the pre-committed SHA-256 MANIFEST, not cryptographic signatures.
- A FAIL is a valid, publishable outcome under this protocol. No parameters, thresholds, or arm definitions were changed after data were seen.

Files

- `ARK_445_preregistration.md/.docx/.pdf` — 28-field preregistration (LOCKED)
- `MANIFEST.txt` — SHA-256 lock manifest
- `RUN_LOG.md` — ordered execution log with commit SHAs
- `selected_qubits.json`, `calibration_snapshot_ibm_marrakesh_20260717.json`
- `spam_results.json` — SPAM kill-gate (PASSED)
- `principal_job_id.txt`, `principal_job_meta.json`
- `raw_results.json` — raw counts (PRIMARY endpoint)
- `proofrecord.json` — full metrics + verdict
- `plots/arm_results.png`, `plots/tristate_discrimination.png`
- Code: `ark_445_select_qubits.py`, `ark_445_circuits.py`, `ark_445_spam_job.py`, `ark_445_submit_ibm.py`, `ark_445_retrieve.py`, `ark_445_analysis.py`

Verdict: FAIL — tri-state discrimination demonstrated ($\Delta_H = 0.463$), but the reset-based confusion/replay DENY arm ($L_conf_corrected = 0.0289$) exceeds the strict 0.02 ceiling. Principal job `d9codk9htsac739c1dug`, SPAM job `d9coda9htsac739c1djg`, `ibm_marrakesh`, 2026-07-17.